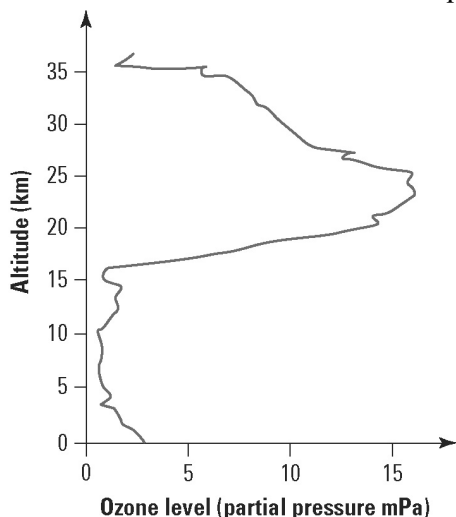


Ozone layer

Student: Class:

Ozone and altitude

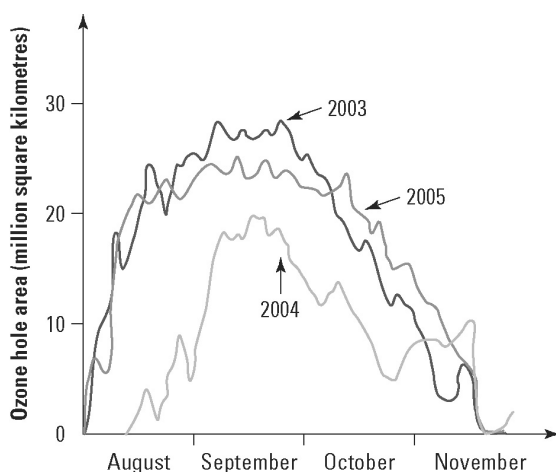
1. The following graph shows the ozone level (measured as the pressure of ozone in millipascals) as a function of altitude. Answer the questions about this graph.



- (a) What is the ozone level at sea level?
- (b) At what altitude is the maximum ozone level?
- (c) In what atmospheric layer is the maximum ozone level located?
.....
- (d) Calculate the ratio of the maximum ozone level in the stratosphere to the maximum ozone level in the troposphere.
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.....

Ozone depletion

2. The following graph shows changes in the area of the ozone hole in the stratosphere above Halley Bay, Antarctica, for the same four-month period over three different years. The ozone hole is a region of the stratosphere that has severely depleted levels of ozone. Answer the questions about this graph.



- (a) In which season of the year do the largest ozone holes appear over Antarctica?
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- (b) Compare the area of the ozone holes over these three years.
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.....
- (c) Determine the size of the ozone hole at the end of September in 2004.
.....
- (d) Studies in Chile showed a significant increase in skin cancers since ozone holes first started to appear in the mid-1980s. Explain why this would happen.
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